

Homework 7

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Problem 1

```
## Warning: package 'gfoRmula' was built under R version 4.4.2
```

```
## Estimated Anticipated Response and Standard Errors:
```

	Treatment	Estimate	SE
## a_0000	a_0000	361.0551	4.699300
## a_0001	a_0001	434.4152	4.843865
## a_0011	a_0011	503.5400	5.667180
## a_0111	a_0111	549.3018	6.589738
## a_1111	a_1111	596.6738	7.758708

Problem 2

Part a

The following logistic regression model gets propensity scores for each year for each patient. The year_round variable is included in the model along with the other covariates because the patients are more likely to have had the transplant as they get later into the study period.

```
##
## Call:
## glm(formula = transplanted ~ year_round + age + sex + edema +
##      serBilir + albumin + prothrombin, family = binomial(), data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.894744   0.493538  -1.813 0.069844 .
## year_round       0.387835   0.016855  23.010 < 2e-16 ***
## age            -0.035689   0.003259 -10.950 < 2e-16 ***
## sexmale         0.337455   0.102449   3.294 0.000988 ***
## edemaedema no diuretics -0.213195   0.160038  -1.332 0.182810
## edemaNo edema    -0.179158   0.150988  -1.187 0.235396
## serBilir        0.005615   0.008287   0.678 0.498065
## albumin        -0.034747   0.075612  -0.460 0.645840
## prothrombin     0.068349   0.028783   2.375 0.017567 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6779.7  on 5698  degrees of freedom
## Residual deviance: 6030.4  on 5690  degrees of freedom
## AIC: 6048.4
```

```
##
## Number of Fisher Scoring iterations: 4
```

Part b

```
## IPW Estimates for Mean Response by Treatment Rule:
```

##	Rule	Estimate
## 1	rule_0	0.03540760
## 2	rule_1	0.02570594
## 3	rule_2	0.21633792
## 4	rule_3	0.18024668
## 5	rule_4	0.21251620
## 6	rule_5	0.44146278
## 7	rule_6	0.34216657
## 8	rule_never	0.12845559

Part c

```
## IPW Estimates with Standard Errors from Bootstrap:
```

##	Rule	Estimate	Standard_Error
## 1	rule_0	0.03540760	0.04015500
## 2	rule_1	0.02570594	0.02057719
## 3	rule_2	0.21633792	0.09714805
## 4	rule_3	0.18024668	0.04248111
## 5	rule_4	0.21251620	0.03505814
## 6	rule_5	0.44146278	0.05339188
## 7	rule_6	0.34216657	0.03851957
## 8	rule_never	0.12845559	0.01714692